POWERBI

Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table:

1. Load both tables:

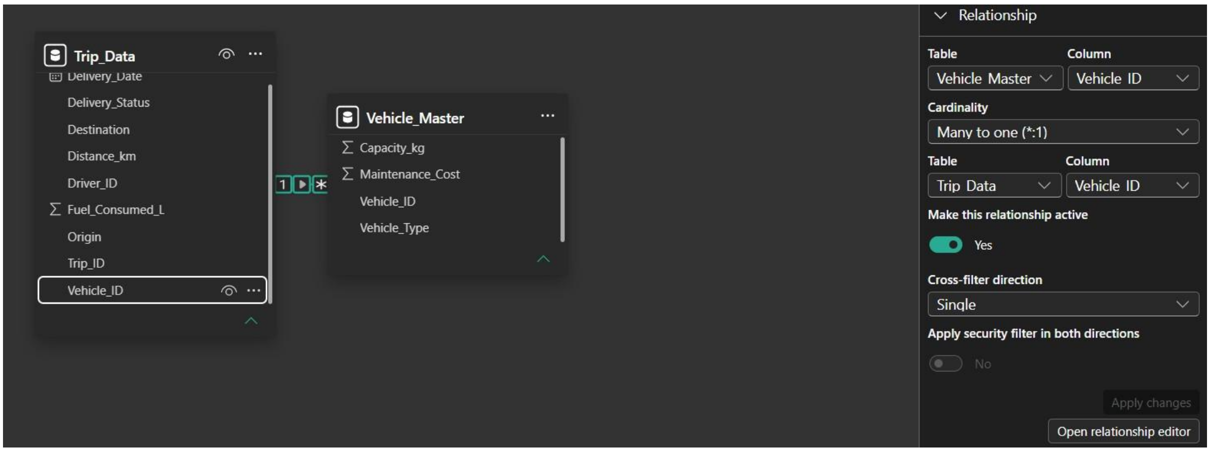
- **Import Trip_Data and Vehicle_Master into Power BI.**

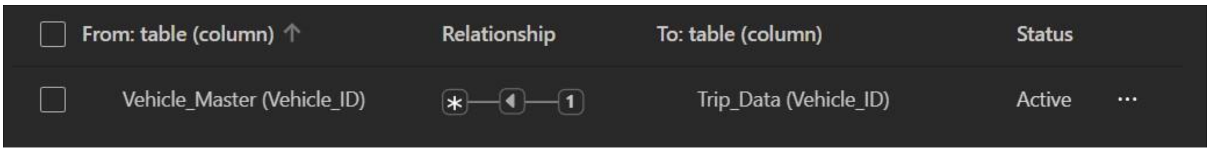
2. Open the Model view:

- **In the left-hand panel, click on the Model icon (it looks like a diagram).**

3. Create the relationship:

- **Drag the Vehicle_ID field from Trip_Data onto the Vehicle_ID field in Vehicle_Master.**
- **Power BI will automatically detect the relationship.**





From: table (column)	Relationship	To: table (column)	Status
Vehicle_Master (Vehicle_ID)	* — 1	Trip_Data (Vehicle_ID)	Active

4. Configure relationship settings:

- **Ensure the relationship is one-to-many.**
- **Cardinality: *One-to-Many (1:*).***
- **Cross filter direction: usually Single.**

DAX Measures:

1. Fuel Efficiency = Distance / Fuel Consumed

```
Fuel Efficiency = DIVIDE(sum(Trip_Data[Distance_km]),sum(Trip_Data[Fuel_Consumed_L]))
```

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel Efficiency
T038	V03	D06	Bangalore	Mumbai	1404	151.36	Late	24 February 2023	10.18
T029	V01	D06	Mumbai	Bangalore	1262	100.72	On-Time	28 January 2023	10.18
T013	V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	10.18
T008	V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	10.18
T007	V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	10.18

2. On-Time Delivery % = On-Time Trips / Total Trips:

To do this, calculate the number of "On-Time" entries in the table and the total entries in the Trip_Data table, and then divide both results.

```
OnTime Count =
CALCULATE(
    COUNTROWS(Trip_Data),
    Trip_Data[Delivery_Status] = "On-Time"
)
```

```
OnTime Delivery % = DIVIDE([OnTime Count], [Total Trips])
```

3. Cost per km = (fuel cost + Maintenance Cost) / Distance

Note fuel cost =100

```
Cost per km =
DIVIDE(
    100,
    SUM(Trip_Data[Distance_km])
)
```

Visualization:

1. Bar chart: On-Time Delivery % by Destination.

2. Line chart: Fuel efficiency trend by delivery date.

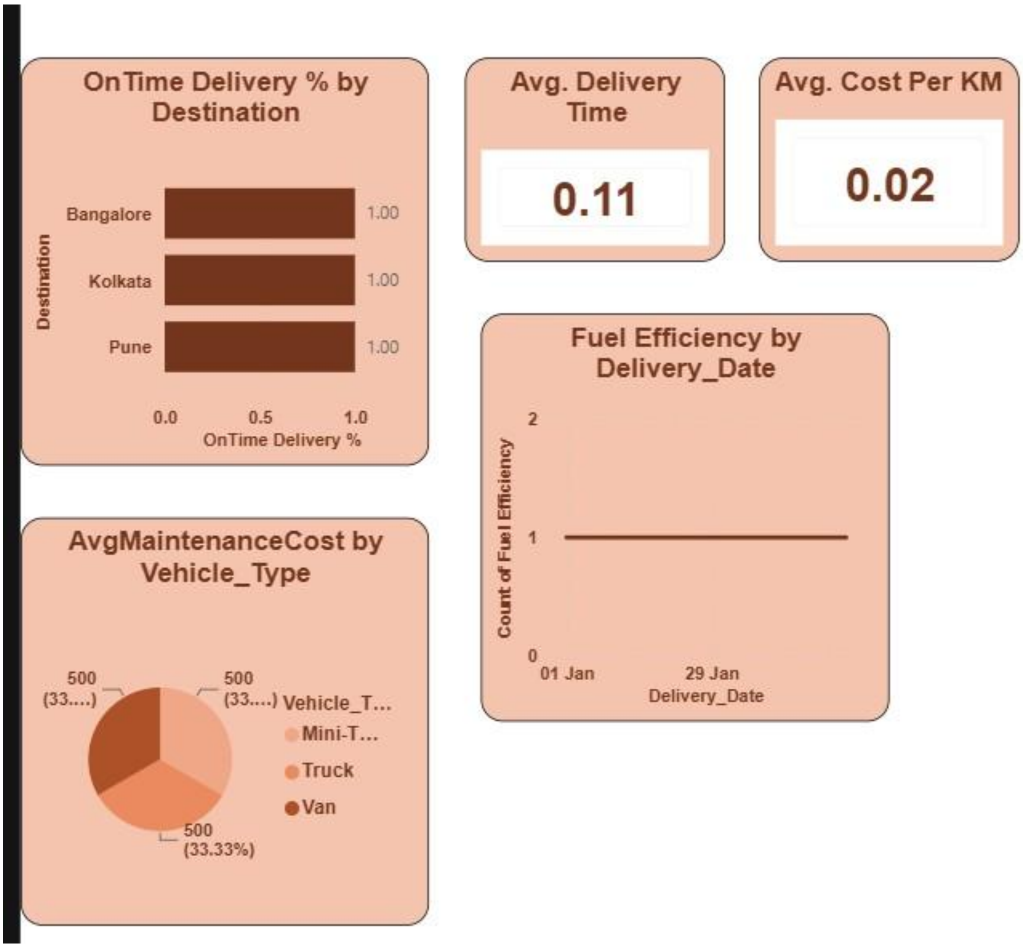
3. Cards visualization:

Avg. Delivery Time

Average cost per km

4. Pie chart:

vehicle type vs Average maintenance cost



AI-Powered Visuals

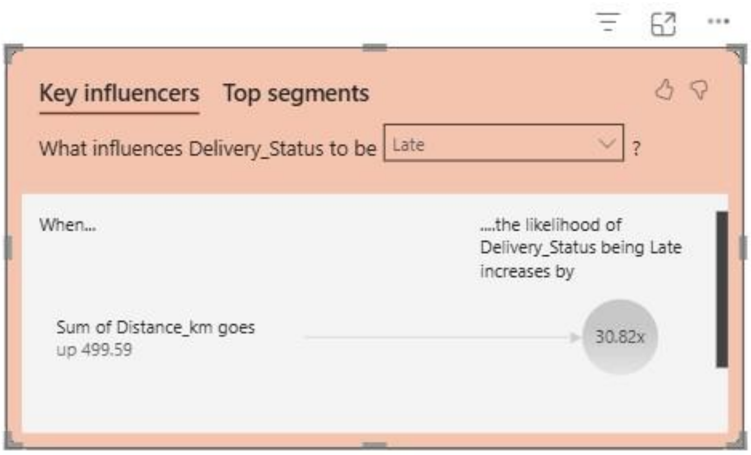
1. **Q&A Visual:**

➢ **Add Q&A visual → Prompt: "Average Cost per km by vehicle type?"**

I don't have Q&A visual. So I Couldn't do it.

2. **Key Influencers Visual:**

➢ **Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID**



I tried this visual somewhat. I couldn't get full information.

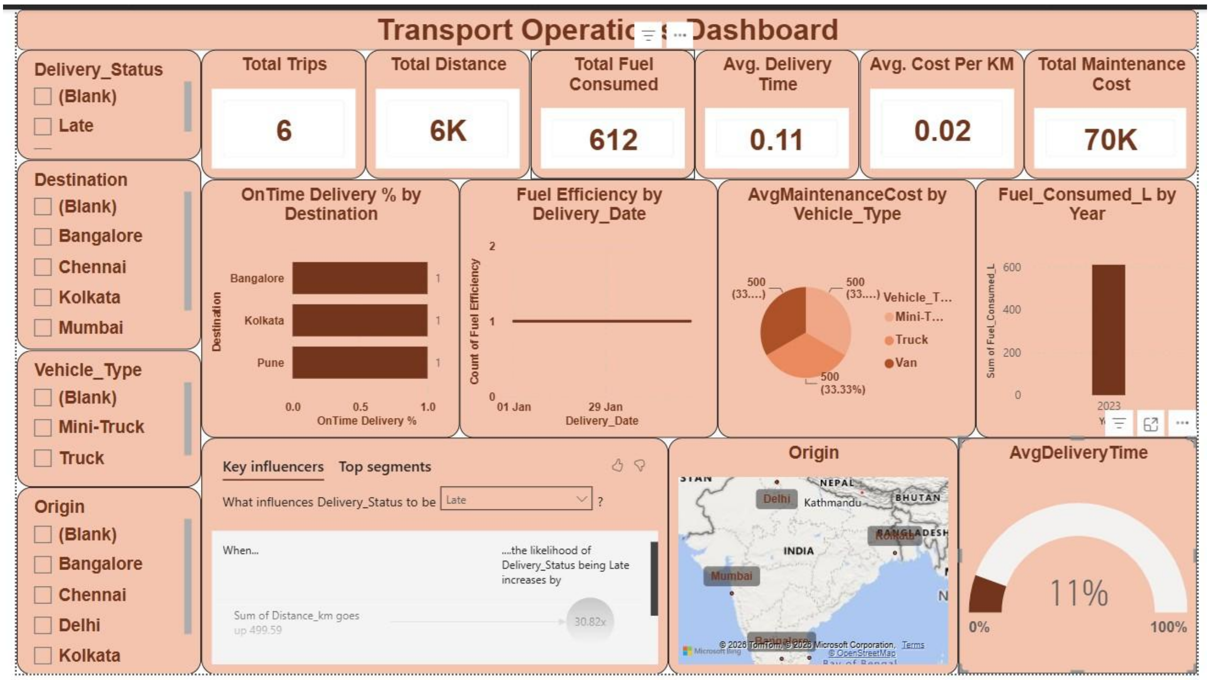
3. Decomposition Tree (AI Visual):

➢ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.

I couldn't try this visual. I couldn't get proper information.

Output:

A transport operations dashboard to optimize routes and fleet usage.



I created Simple Dashboard to view simple transport operation details.

I used Cards to view Avg & Total for Quick Analysis.

I couldn't predict Q&A Analysis, Key influencers & Decomposition Tree.